

Bocconi

COMPUTER SCIENCE

code 30424 a.y. 2024-2025

Lesson 5

Highlighting and visualizing data



Università
Bocconi
MILANO

Objectives of the lesson

Understand how to use conditional formatting to automatically highlight specific information in a table and how to visualize data effectively using Excel charts

Our first case

A company needs to purchase solar panels to improve energy efficiency. In the **Scores** worksheet are the scores of 12 key features for 6 solar panel models, based on expert evaluations. We need to:

- Use a function to find a single score for each solar panel model, calculating a weighted average of the key features scores
- Apply a format to the total score cells, so that the values corresponding to the highest and lowest scores are automatically highlighted with two fill colors of our choice
- Apply a “3 traffic lights” format to the total score cells, so that the highest score has a green icon and the lowest score has a red icon
- Apply a “Data bar” format to the scores values of the solar panel model with the greatest total score
- Apply a format to the key features labels so that they are automatically highlighted with a fill color of our choice when the weight is greater than or equal to 10%

Our second case

In the **Solar panels** worksheet there's a list of solar panels sold by a European wholesaler. We need to:

- Calculate, in a new column named **Final price**, the final price of each solar panel, considering the discount
- Apply a format to the **Producer** column cells so that they are automatically highlighted with a fill color of our choice when corresponding to models with no discount and with shipping costs not available or currently being updated
- Calculate, in a new column named **Ranking**, the rank of each final price so that the highest rank corresponds to the lowest value
- Apply a format to the cells of the **Model** column so that the values are automatically highlighted with a fill color of our choice when the solar panel is made in USA or Japan and the model's rank is one of the first 15

Our third case

In the file we have the **Homes** worksheet we have historical rental prices for five homes and the **Company** worksheet with the number of employees and the total revenues for a company for a 10 years period. We need to:

- Create a chart in the **Homes** worksheet representing the rental price of Home 2 in the period 2017-2024. Then we need to:
 - add a linear trendline to the chart
 - add to the chart also the Home 5 rental price series with its own trendline
- Create a chart in the **Company** worksheet representing the employees and the revenues for the 2015-2024 period. Then we need to :
 - add a title for both vertical axes
 - show the revenues in thousands with no decimal places

Tools to learn with today's lecture

- Conditional formatting
- Conditional formatting with a formula
- IS functions
- RANK.EQ function
- Charts with a trendline, mixed and with two axes

Book references

Excel for students in economics and finance, 2nd edition:

Chapters 1.3, 1.5, 3.2.5, 4.2

Excel workbook, 4th edition:

Exercises on the functions and tools of the lesson